

Upendra Dommaraju

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Professional Summary

Full-Stack Engineer with AI integration expertise, delivering cloud-native services at a senior level. Designed and deployed Kafka-driven event workflows and Redis cache-aside patterns, boosting processing throughput and API performance under peak load. Proven track record in microservices architecture, Node.js, and performance optimization to build scalable, reliable systems. Seeking to leverage this experience to accelerate product innovation and drive business growth.

Technical Skills

- **Programming:** Java, JavaScript (ES6+), TypeScript, Python, SQL
- **Full Stack & API Engineering:** HTML5, CSS3, React.js, Redux, Spring Boot, Node.js, Express.js, Next.js, Angular, RESTful API Design-, API Versioning, Idempotency, Authentication & Authorization (JWT, OAuth 2.0), Asynchronous Programming, API Documentation, Secure API Design, Role-Based Access Control (RBAC)
- **AI Integration & LLM Engineering:** LLM API Integration, Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings & Semantic Search, Vector Databases, AI Workflow Orchestration, Streaming AI Responses, AI Cost Optimization, Secure & Scalable AI System Design, Retry & Fallback Strategies, Prompt Injection Mitigation
- **Microservices & Distributed Systems:** Microservices Architecture, Event-Driven Systems, Asynchronous Messaging, Service-to-Service Communication, System Design, Apache Kafka, API Gateway, API Testing
- **Databases & Caching:** MongoDB, PostgreSQL/MySQL, Redis, Indexing Strategies, Data Modeling, Read/Write Optimization, Consistency & Performance Trade-offs
- **Cloud-Native & Platform Engineering:** CI/CD, Docker, Kubernetes (Pods, Services, Deployments, Autoscaling), AWS (EC2, S3, RDS, Lambda), Cloud-Native Design Principles, Nginx (Reverse Proxy, Load Balancing)
- **Observability & Operations:** Structured Logging, Metrics (Prometheus-style), Monitoring, Alerting, Production Debugging
- **Core Computer Science:** Data Structures & Algorithms, Object-Oriented Design, Operating Systems, Database management systems

Experience

CoverMyMeds

Jun 2025 - Present

Software Engineer

Twinsburg, Ohio

- Delivered scalable backend services with Node.js, Express.js, and MongoDB on AWS EC2, enabling reliable processing of high-demand workflows, improving system stability and reducing downtime
- Helped evolve the platform from tightly coupled services to a MERN microservices architecture using Docker containers and AWS Lambda, increasing deployment independence, improving fault isolation, and speeding release cycles
- Designed and integrated RESTful APIs and maintained legacy SOAP integrations as needed, enabling secure communication with internal services and external partners and reducing integration errors
- Worked with frontend teams to deliver features using React.js, aligning UI components, API contracts, and backend logic, which enabled on-time feature releases and improved UI consistency
- Built MongoDB-backed data processing pipelines using Node.js and modern JavaScript patterns (async/await, array methods), supporting both batch and transactional workloads while significantly reducing processing time.
- Integrated LLM APIs into backend workflows of MERN applications to automate text analysis, content generation, and decision support, reducing manual review time and improving response consistency
- Designed backend pipelines that use prompt engineering and structured JSON outputs with workflow orchestration, enabling reliable and scalable AI integrations that increase processing throughput and simplify maintenance

Accenture

Jan 2020 - Nov 2023

Full Stack Engineer

Chennai, India

- Owned the design and evolution of Node.js and Express.js backend services, enforcing clear controller-service-data layer boundaries to improve code quality, testability, and long-term maintainability across MERN applications.
- Designed and implemented Kafka-based event-driven workflows to support asynchronous processing, improving system resilience and reducing tight coupling between MERN microservices.
- Led MongoDB data modeling and indexing strategies, aligning schemas with real access patterns to ensure consistent low-latency performance under high-throughput workloads.
- Introduced a Redis cache-aside pattern for hot-read API paths, offloading MongoDB and improving average API response times by up to 40% during peak traffic.
- Designed and secured RESTful APIs using Express.js with authentication, authorization, and request validation, enabling reliable frontend consumption from React-based user interfaces.
- Collaborated with frontend engineers to define stable API contracts and data shapes using OpenAPI specifications, enabling React components to consume Node.js services and reducing integration defects
- Drove containerization (Docker) and Kubernetes deployment standards for MERN services, improving deployment consistency, horizontal scalability, and operational visibility across environments.

Education

University of Dayton

Jan 2024 - Dec 2025

Master of Science, Computer Science

Dayton - Ohio

- **GPA:** 3.7